Dedicated Mentoring System for Students (HEPro AI+)

Project Title: Dedicated Mentoring System for Students (HEPro AI+)

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Abstract

Educational institutions increasingly face challenges in providing personalized mentoring at scale. Traditional mentoring systems are largely manual and reactive, making early identification of at-risk students difficult. Modern students require support across multiple dimensions including academics, wellness, productivity, and career readiness.

This project presents the HEPro AI+ Dedicated Mentoring System, a hybrid intelligence framework combining rule-based scoring and machine learning. The

system processes structured student data to compute performance scores, derive a Student Readiness Index (SRI), segment students using K-Means clustering, and automatically assign mentors based on expertise and availability.

The solution is implemented using Python, pandas, and scikit-learn. Results demonstrate effective student segmentation, interpretable risk detection, and balanced mentor allocation. The proposed system provides a scalable foundation for AI-driven student mentoring in modern educational environments.

1. Introduction

1.1 Background

Student success in modern education depends on multiple factors including academic performance, mental wellbeing, productivity habits, and career preparedness. However, most traditional mentoring systems rely heavily on manual monitoring and periodic faculty reviews. This makes early risk detection difficult and often delays intervention for students who silently struggle.

With the growing availability of educational data, institutions now have an opportunity to transform mentoring through data-driven intelligence. Artificial intelligence techniques can continuously monitor student performance across multiple dimensions and provide actionable insights to mentors.

The HEPro AI+ Dedicated Mentoring System is developed to address this need. It integrates academic, wellness, productivity, and career indicators into a unified analytical pipeline that supports proactive and scalable mentoring decisions.

1.2 Motivation

Many students experience high stress, poor time management, or lack of career clarity even when their GPA appears acceptable. Traditional mentoring systems typically detect problems only after academic decline becomes visible, which limits the effectiveness of intervention.

Another key motivation is scalability. Mentors often handle large numbers of students and cannot manually track multi-dimensional performance indicators. An intelligent support system is required to automatically highlight at-risk students and prioritize mentor attention.

The project is also motivated by the importance of interpretability in educational AI systems. Pure black-box models may not be easily trusted by academic stakeholders. Therefore, a hybrid approach combining rule-based scoring with machine learning was chosen to balance transparency and analytical power.

1.3 Problem Definition

Existing mentoring systems suffer from several limitations:

- Over-reliance on GPA and attendance
- Lack of holistic student profiling
- Delayed identification of at-risk students
- Manual and inefficient mentor allocation
- Limited scalability across large student populations

The core problem addressed in this project is how to design an interpretable and scalable hybrid AI system that integrates multi-dimensional student data to proactively identify risk groups, segment students meaningfully, and recommend targeted mentoring actions.

1.4 Objectives of the Project

The primary objectives of this project are:

- To design a realistic student mentoring dataset
- To develop rule-based performance scoring metrics
- To compute a holistic Student Readiness Index (SRI)
- To apply K-Means clustering for student segmentation
- To build a mentor-student matching mechanism
- To generate actionable intervention recommendations
- To simulate an end-to-end mentoring intelligence pipeline

1.5 Scope of the Project

The system focuses on the intelligence layer of student mentoring, including data preprocessing, scoring, clustering, mentor matching, and alert generation. The project is implemented as a prototype designed for future institutional scaling.

Real-time deployment and integration with live institutional databases are outside the current scope.

1.6 Limitations

The system has the following limitations:

- Synthetic dataset used instead of real institutional data
- Limited sample size of students
- Static mentor availability assumptions
- K-Means clustering assumes spherical cluster structure
- No real-time dashboard implementation

Despite these limitations, the system demonstrates strong conceptual validity and practical potential.

2. Literature Review / Conceptual Foundation

Traditional mentoring systems are primarily manual and reactive, relying on periodic faculty reviews and subjective judgment. Such systems often fail to detect early behavioral risks and do not scale effectively across large student populations.

Recent advances in artificial intelligence have enabled predictive analytics, personalized learning systems, and automated academic advising. Research indicates that combining academic indicators with behavioral and wellness data provides significantly better insight into student risk than GPA alone.

Hybrid AI systems that combine rule-based reasoning with machine learning have gained attention because they balance interpretability with pattern discovery. Pure rule-based systems lack adaptability, while purely data-driven models may lack transparency. This project adopts a hybrid architecture to ensure both usability and analytical depth.

3. System Overview

3.1 Project Architecture

The HEPro AI+ system follows a layered architecture:

Data Layer → Intelligence Layer → Decision Layer → Action Layer → Feedback Layer

This modular design improves clarity, scalability, and maintainability of the system.

3.2 System Workflow

The overall workflow of the system is as follows:

1. Student data collection
2. Data preprocessing and normalization
3. Rule-based score computation
4. Student clustering using K-Means
5. Risk classification
6. Mentor matching
7. Intervention recommendation
8. Alert generation

3.3 Module Breakdown

Data Layer: Handles student and mentor datasets.

Intelligence Layer: Performs scoring and clustering.

Decision Layer: Identifies risk levels and primary student needs.

Action Layer: Assigns mentors and generates interventions.

Feedback Layer: Produces alerts and monitoring outputs.

4. Dataset Design

4.1 Dataset Philosophy

The dataset was synthetically generated to simulate realistic student behavior across academics, wellness, productivity, and career readiness. Logical

relationships were preserved, such as higher stress correlating with lower wellbeing and productivity.

4.2 Student Dataset Structure

The student dataset includes demographic, academic, behavioral, and engagement features. Key attributes include GPA, attendance, assignment completion, stress level, sleep hours, mental wellbeing, productivity score, distractions, career clarity, skill readiness, and engagement score.

From these base attributes, composite indicators (APS, WWS, PTMS, CRS, and SRI) were derived to provide interpretable metrics for mentoring decisions.

4.3 Mentor Dataset Structure

The mentor dataset includes mentor ID, mentor name, expertise domain (Academic, Wellness, Career), maximum student capacity, current workload, and availability level. This structure enables realistic mentor allocation based on both expertise matching and workload balancing.

4.4 Data Preprocessing

Data preprocessing involved validation of value ranges, selection of relevant numerical features, and normalization using StandardScaler. These steps ensured that features with different ranges did not bias the clustering algorithm.

4.5 Data Quality Controls

Quality checks included range validation, logical consistency verification, and inspection of feature distributions. These controls ensured that the synthetic dataset remained realistic and analytically useful.

5. Feature Engineering

Relevant features were selected to capture multi-dimensional student performance. Standard normalization was applied before clustering to ensure fair distance computation.

Derived composite scores were engineered using weighted formulas:

- Academic Performance Score (APS)
- Wellness and Wellbeing Score (WWS)

- Productivity and Time Management Score (PTMS)
- Career Readiness Score (CRS)
- Student Readiness Index (SRI)

These features improved interpretability and enabled holistic student evaluation.

6. Scoring System Design

A rule-based intelligence framework was developed to convert raw student attributes into meaningful performance indicators.

APS measures academic consistency using GPA, attendance, and assignment completion.

WWS captures mental and physical wellbeing using mental wellbeing score, sleep hours, and inverse stress.

PTMS evaluates time management using productivity score, inverse distractions, and attendance.

CRS measures career preparedness using career clarity, skill readiness, and engagement.

SRI provides a holistic readiness indicator by combining all component scores.

Students were classified into Green, Blue, Yellow, and Red categories based on SRI thresholds to support mentor prioritization.

7. Machine Learning Design

K-Means clustering was applied to discover hidden student segments. Prior to clustering, features were standardized to ensure fair distance computation.

The model identified four major behavioral profiles:

- High Performers
- At-Risk Students
- Career-Confused Students
- Balanced Students

Comparison between clustering results and SRI categories showed strong alignment, validating the hybrid intelligence approach.

7.3 Mentor Matching Logic

Students were mapped to mentors using a rule-based need detection mechanism. Priority was given to wellness risks, followed by academic weakness and career uncertainty.

Mentors were filtered based on expertise and available capacity. The system assigned the least-loaded qualified mentor to ensure balanced workload distribution.

8. Intervention and Decision System

The decision engine translates analytics into actionable mentoring steps. Students with low wellness scores or high stress are assigned wellness counseling. Students with weak academic indicators receive academic support, while students with low career readiness receive career guidance.

Red-category students trigger urgent alerts for immediate mentor attention. Yellow-category students are flagged for close monitoring. This ensures proactive intervention and efficient mentor resource utilization.

9. Mentor Playbook Framework

Structured playbooks were defined for each student category to standardize mentor actions. The playbooks specify recommended intervention type, follow-up frequency, and monitoring strategy. This framework ensures consistency and scalability in mentoring operations.

10. System Implementation

The system was implemented using Python-based tools including pandas for data processing, scikit-learn for machine learning, and matplotlib/seaborn for visualization.

The processing pipeline follows:

CSV Input → Preprocessing → Scoring → Clustering → Need Detection → Mentor Matching → Output Generation

The modular notebook design supports easy extension and institutional scaling.

11. Results and Outputs

The HEPro AI+ system successfully generated interpretable readiness scores, identified hidden student clusters, detected at-risk students, and assigned mentors efficiently.

The integration of rule-based intelligence with machine learning improved both transparency and analytical depth. The system demonstrated the feasibility of scalable, data-driven mentoring support.

12. Evaluation

Performance: The system effectively differentiated student readiness levels.

Interpretability: High, due to transparent rule-based scoring.

Scalability: Modular architecture supports expansion.

Reliability: Logical consistency observed across outputs.

13. Discussion

Key learnings include the effectiveness of hybrid AI for educational analytics and the importance of combining behavioral and academic indicators. Challenges included designing realistic synthetic data and tuning score weights. Interpretability was prioritized over black-box complexity to maintain mentor trust.

14. Ethical and Social Considerations

Student data was anonymized using unique identifiers to preserve privacy. Feature selection aimed to minimize bias across student groups. The system follows a human-in-the-loop philosophy where mentors remain the final decision makers. Responsible AI principles of transparency, fairness, and accountability were emphasized.

15. Future Scope

Future enhancements may include:

- Integration with real institutional databases
- Predictive dropout modeling
- Real-time mentoring dashboards
- Advanced ML recommendation systems
- Large language model based mentor assistants

16. Conclusion

The HEPro AI+ Dedicated Mentoring System demonstrates how hybrid AI can transform student support into a proactive and scalable process. By combining rule-based scoring, machine learning segmentation, and automated mentor matching, the system provides meaningful and actionable mentoring intelligence.

The framework forms a strong foundation for future deployment in educational institutions seeking data-driven student support solutions.